

Timothy Baird

Software Developer · Educator · Instructional Designer

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PROFESSIONAL SUMMARY

Practitioner-educator with over a decade of experience teaching software development at VET and higher education level — combined with active, ongoing development work across web, blockchain, AI/LLM, and database systems. Proven ability to bridge the gap between industry practice and structured learning, with a strong track record in instructional design, curriculum development, and regulatory compliance. Brings genuine technical depth to every teaching engagement, and genuine pedagogical skill to every technical project.

RECENT PROJECTS & TECHNICAL WORK

Full Circle MUD · [Solo Developer](#)

Ongoing

A multiplayer online game (MUD) built on the Evennia/Django engine, with XRPL blockchain and AI/LLM integration. One major goal has been to engineer a genuinely decoupled system — distinct game subsystems interact only through well-defined interfaces, enforced via bespoke Python libraries.

Key technical work:

- XRPL on-chain integration — supporting APIs for multisig transactions and NFT metadata
- LLM integration via OpenRouter and OpenAI APIs, including creation and consumption of embeddings stored in vector databases
- Horizontal scaling framework: engineered a solution to distribute Evennia's single-threaded Twisted Reactor across multiple instances sharing a single vertically-scaled Postgres instance
- XRPL wallet management interface (private repo) for internal operational use
- LLM test-bot harness (private repo) for automated game dialogue testing
- Python library development with a focus on minimal coupling between game systems
- [github.com/fullcirclemud](https://github.com/timbaird/fullcirclemud)
- Live: fcmud.world

Diploma of IT — Cloud & Cyber Security · [Instructional Design / Development](#)

Ongoing

End-to-end development of course content and assessments for a Diploma of Information Technology (Cloud & Cyber Security) at Bendigo Kangan Institute, including a supporting website built with Astro.

- Full curriculum and assessment design from scratch
- Supporting website: github.com/Digital-Initiatives-Kangan-Institute/diploma-cloud-cyber-website
- Course repository: github.com/Digital-Initiatives-Kangan-Institute/diploma-cloud-cyber

Personal Projects · github.com/timbaird

Ongoing

- Google Calendar API integration for use with Go High Level CRM platform
- AI/LLM learning exercises and code samples developed for student use
- Additional private projects spanning Oracle PL/SQL, C#, Solidity (smart contracts), and JavaScript

WORK EXPERIENCE

VET Teacher — IT & Software Development · [Bendigo Kangan Institute](#) 2025 – 2026

Responsibilities:

- Preparing and delivering classes across IT and software development disciplines
- Student assessment and feedback
- Instructional design — developing course structure, content, and assessments
- Compliance with ASQA industry regulation and standards

Disciplines covered:

- Database design, build, and deployment
- Software development processes — CI/CD and SCRUM
- Design Thinking and Double Diamond Framework
- Authentication and security fundamentals
- Cloud theory and AWS services

Instructional Designer · [Bendigo Kangan Institute \(Contractor\)](#) 2023 – 2024

Worked within a small contractor team to plan and develop learning materials for Software Development and Data Analysis VET courses and micro-credentials.

- Developed learner journey concept and strategy
- Created learning and assessment materials, course artifacts, and related documentation

Teacher — IT & Software Development · [Swinburne University of Technology](#) 2015 – 2022

Primary teacher for Certificate IV in Information Technology (Programming) and Diploma of Software Development. One of the longer-tenured practitioners in the department, responsible for full lifecycle of teaching, assessment, and industry engagement.

Responsibilities:

- Planning, preparing, delivering, and assessing across multiple VET subjects
- ASQA compliance — ensuring training and assessment met all regulatory requirements
- Student engagement evidence collection for government funding compliance
- Collaborating with teaching team to deliver a cohesive course experience across subjects
- Engaging with industry partners to ensure currency and relevance of skills being taught

Technologies & skills taught:

- Database design and programming — MS SQL Server / T-SQL and Oracle / PL/SQL
- Object-oriented and procedural programming — primarily C# (.NET), with Python and JavaScript
- Layered, loosely-coupled and decoupled architecture patterns
- Unit testing and client acceptance testing
- RESTful API development and consumption (.NET environment)
- Cloud deployment and management — AWS and Azure
- CI/CD pipelines, version control (Git)
- Algorithmic problem solving
- Project management — modified SCRUM and Waterfall methodologies

Tutor / Casual Teacher · [Swinburne University of Technology](#) 2014 – 2015

Taught at both higher education (Bachelor) and vocational (TAFE) level.

- Delivering tutorials and classes
- Conducting and grading assessments
- Supporting students with technical problems

Data Analysis & Reporting Specialist · [Accenture](#) 2013 – 2014

Employed on an Accenture contract embedded with one of Australia's Big 4 banks.

- Sourcing, processing, and storing data from across the business areas covered by the contract
- Analysing data and reporting to both Accenture and client leadership on contract KPI compliance
- Identifying inefficiencies and opportunities for improvement in business processes

Technical Support — Apple Environments · Mac Aid

2007 – 2012

Help desk and onsite technical support across SME business and consumer environments, primarily Apple hardware and ecosystems, including integration with network, server, SAN, and associated services. Windows and Linux environments as required.

Personal Trainer · Self Employed

2006 – 2007

Leading individual and group fitness sessions across multiple gyms and studios.

Combat Systems Operator / Weapons Performance Analyst · Australian Defence Force — Royal Australian Navy

2000 – 2006

Served at HMAS Cerberus, HMAS Watson, HMAS Sydney, HMAS Kuttabul, and HMAS Success.

- Roles including Trainee, Combat Systems Operator, Ships Boarding Party, Weapons Performance Analyst, and Heavy Vehicle Driver

TECHNICAL SKILLS

Languages	Python, C# (.NET), JavaScript, SQL (T-SQL, PL/SQL), Solidity, Bash
Frameworks	Django / Evennia, ASP.NET, Astro
AI / LLM	OpenAI API, OpenRouter, vector databases, embeddings
Blockchain	XRPL, multisig, NFT metadata, EVM smart contracts (Solidity)
Cloud & DevOps	AWS, Azure, CI/CD pipelines, Git
Databases	PostgreSQL, MS SQL Server, Oracle, vector stores
Architecture	Loosely-coupled / decoupled systems, RESTful APIs, layered design
Teaching & Design	VET instructional design, ASQA compliance, curriculum development, assessment design

EDUCATION

Bachelor of Computer Science (Honours) · Swinburne University of Technology	2018
Diploma of Vocational Education & Training · Swinburne University of Technology	2017
Bachelor of Information & Communication Technology · Swinburne University of Technology	2013
Advanced Diploma of Computer Science · Swinburne University of Technology	2011

AWARDS & RECOGNITION

- PAVE Excellence Award for Inclusion and Diversity — 2019
- Most Outstanding First Year Student — Swinburne University of Technology, 2011
- Australian Active Service Medal
- Afghanistan Campaign Medal
- Rotary Youth Leadership Award
- Queen's Scout Award

INTERESTS

Basketball. Snow skiing. Settlers of Catan. Tabletop RPGs. AI. Crypto - General geek stuff.

REFEREES

Andrew Roadknight

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Currently Bendigo Kangan Institute
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